

# Amutheezan Sivagnanam

U.S. Permanent Resident | [Scholar](#) | <https://www.linkedin.com/in/amutheezansivagnanam> | [Github](#) | Open to relocation

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## SUMMARY

- PhD Candidate and Machine Learning Researcher specializing in Deep Reinforcement Learning
  - Six years of academic research experience building scalable reinforcement learning and deep learning systems for efficient ML training and deployment
  - Two years of industrial experience in software development, behavioral driven testing and CI/CD while following agile practices
  - Good publication record with an h-index of 6 and 100+ citations, featuring in top-tier AI and ML conferences
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## EDUCATION

**Ph.D. Computer Science**, (August 2025): **Pennsylvania State University**, University Park, PA, Advisor: Dr. Aron Laszka

- **Dissertation**: Application of Deep Reinforcement Learning to Solve Optimization Problems in Transportation Domains

**M.S. Computer Science**, (August 2022): **University of Houston**, Houston, TX, Advisor: Dr. Aron Laszka

## RESEARCH EXPERIENCE

**Graduate Research Assistant**, *Pennsylvania State University*, (August 2022 - May 2025)

- Devised and evaluated a **Transformer based multi-agent reinforcement learning algorithm** to solve the **emergency responder reallocation problem** that **reduces decision making time by 3 orders of magnitude**
- Leveraged a **deep reinforcement learning** approach to tackle the challenge of online vehicle routing with advance booking, enabling **prompt confirmation within seconds**

**Graduate Research Assistant**, *University of Houston*, (September 2019 - August 2022)

- Introduced a novel **reinforcement learning** based solution approach to solve the problem of **online booking for offline vehicle routing problem** which resulted in **20 % reduction of operational costs** for the transit agency
- Introduced **novel problem formulation and algorithms** that **lower the energy costs by \$140k** annually for public transit agencies operating mixed fleets of buses
- **Skills**: Python, C++, Java, C, SQL, MySQL, MongoDB, SQLite, PyTorch, TensorFlow, Scikit-learn, Transformers, Machine Learning, Deep Learning, Computer Vision, Reinforcement Learning, DQN, DDPG, TRPO, PPO, GRPO, LLM, RLHF, RAG, Prompt Engineering, LangChain, Bayesian Optimization, Data Science, Data Visualization, Clustering, Statistical Analysis, Data Mining, SLURM, Ray, RLLib, OpenAI Gym, NumPy, pandas, matplotlib, Linear Programming, Integer Programming, CPLEX, Gurobi, Mosek, OR-Tools, Spark, Amazon Web Services (Sage Maker, S3, Lambda), Git, Docker, Linux, Bash, Object-Oriented Development, MATLAB

## SELECT PUBLICATIONS

A. Sivagnanam et al.; Multi-Agent Reinforcement Learning with Hierarchical Coordination for Emergency Responder Stationing. International Conference on Machine Learning. ICML 2024. **Core Ranking A\***

A. Sivagnanam et al.; Offline Vehicle Routing Problem with Online Bookings: A Novel Problem Formulation with Applications to Paratransit. International Joint Conference on Artificial Intelligence. IJCAI 2022. **Core Ranking A\***, Invited Talk: Vienna Austria

A. Sivagnanam et al.; Minimizing Energy Use of Mixed-Fleet Public Transit for Fixed-Route Service. AAAI Conference on Artificial Intelligence. AAAI 2021. **Core Ranking A\***, Invited Talk: Virtual Conference

## SOFTWARE ENGINEERING EXPERIENCE

**Software Engineer**, *LSEG Technology*, (January 2018 - July 2019)

- Introduced **unit testing** for libraries in the Post-Trade C++ codebase, improving system performance and reliability
- Implemented and validated **Oracle database changes** for Post-Trade products for the Singapore Stock Exchange using **Behavior Driven Development (BDD)** testing approaches in **Java**, ensuring data integrity and system functionality
- Contributed to the **CI/CD pipeline** of the Post-Trade product using **Python and Git** ensuring seamless software updates

**Software Engineering Intern**, *WSO2 Lanka PVT Ltd*, (July 2016 - December 2016)

- Engineered an **alert generation system** to monitor disease outbreaks by analyzing descriptive **HL7/FHIR data**, automating email and SMS notifications, and supporting rapid decision-making through prompt data processing
- Developed a mechanism to evaluate hospital functionality by assessing bed and oxygen cylinder availability from admission and discharge messages, with **effective requirements gathering** and integration of **structured data processing**
- **Skills**: Python, C++, JAVA, JS, PHP, SQL, Oracle DB, Object Oriented Development, Github, Linux, Agile Practices, Behaviour Driven Development, Selenium, Vaadin, Natural Language Processing, Data Science, Data Visualization, HL7, FHIR, CI/CD

## REFERENCES

|                    |                                                    |                                                          |                       |
|--------------------|----------------------------------------------------|----------------------------------------------------------|-----------------------|
| <b>Name</b>        | Dr. Aron Laszka (PhD Advisor)                      | Dr. Abhishek Dubey                                       | Dr. Weidong Shi       |
| <b>Position</b>    | Assistant Professor                                | Associate Professor                                      | Associate Professor   |
| <b>Affiliation</b> | Pennsylvania State University                      | Vanderbilt University                                    | University of Houston |
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